

Load flow: DG connected vs base case

IEEE 13-node feeder - all buses and branches, From node -> To node

Base case: DG disconnected. **Compared case:** 4.05 MVA synchronous DG connected at node 692 through its 0.69/4.16 kV transformer, dispatched at 3240 kW. Both are AC unbalanced three-phase load flows in PowerFactory (project *IEEE13 Yousaf2022 Replication*, study case *Study with Substation Transformer*, regulator taps fixed). No relay, fuse or network setting differs between the two cases.

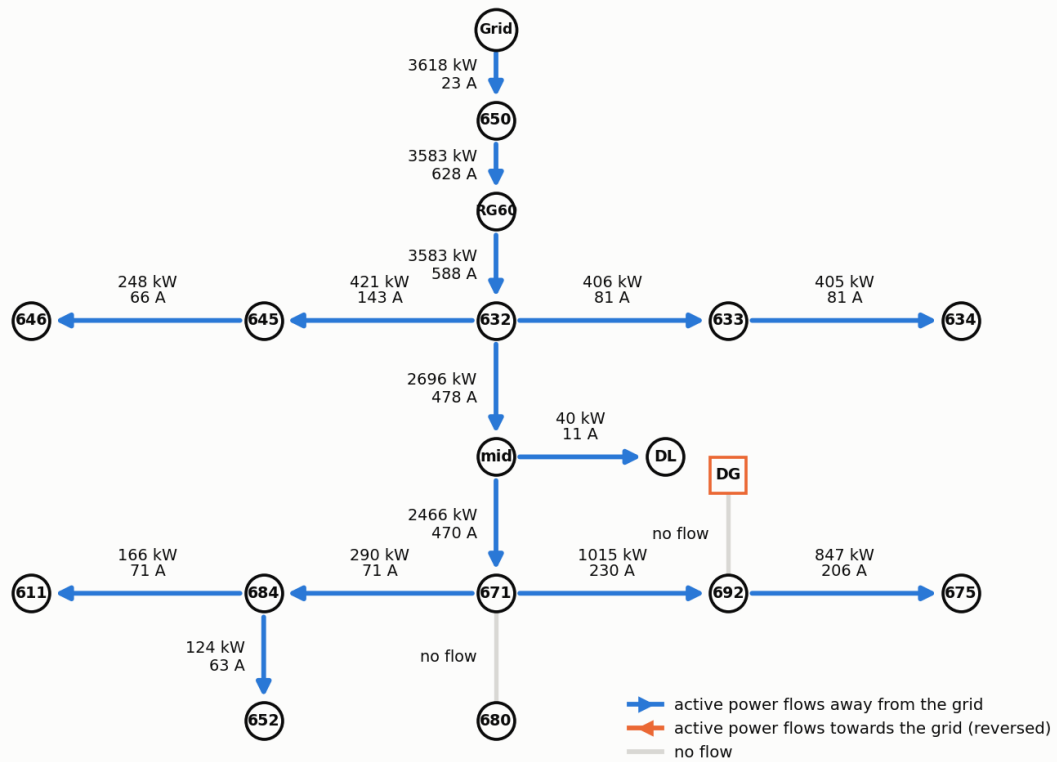
Every branch is written **From node -> To node** in the direction away from the grid. Values are taken at the From node: + means the power or current flows From -> To, - means it flows To -> From (towards the grid). The power balance closes to 0.03 kW at every bus in both cases.

Summary

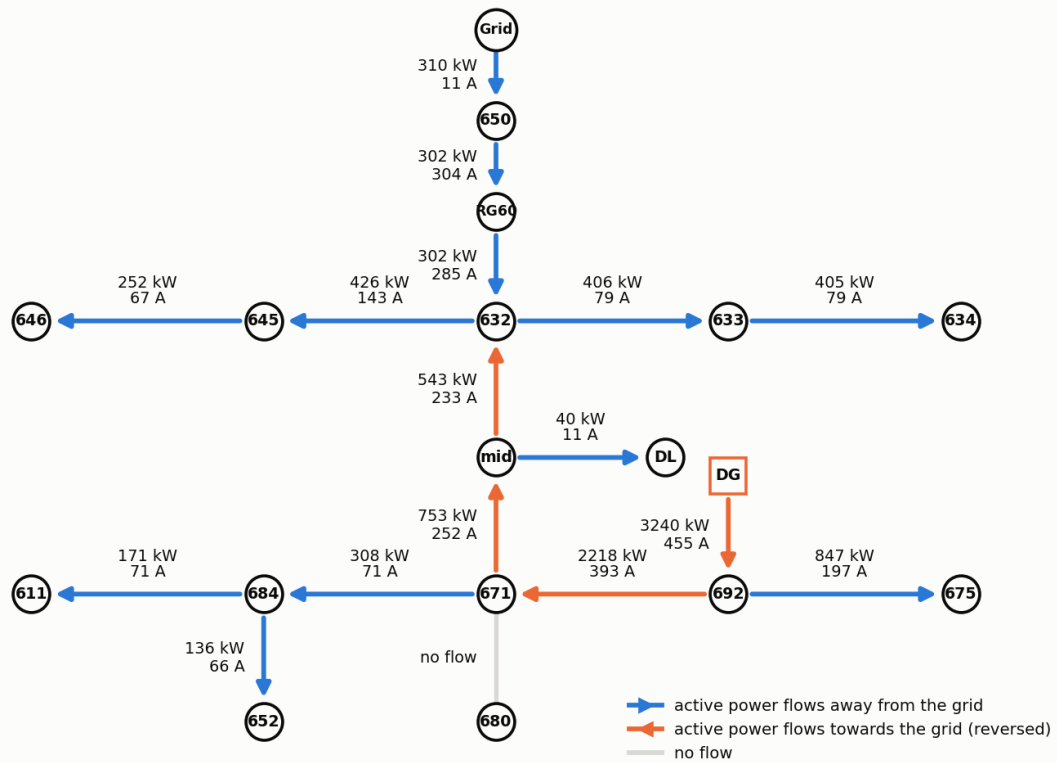
Quantity	Base case (DG out)	DG connected	Change
Power from the grid	3618.2 kW / 2003.3 kvar	310.3 kW / 1900.6 kvar	-3307.9 kW (-91 %)
Power from the DG	-	3240.0 kW / 0.0 kvar	
Total load	3471.1 kW / 2104.3 kvar	3500.6 kW / 2124.0 kvar	+29.5 kW
Capacitors supply	707.9 kvar	747.4 kvar	+39.5 kvar
Total losses	147.1 kW	49.6 kW	-97.5 kW (-66 %)
Lowest feeder voltage	0.9712 pu (611 phase c)	1.0055 pu (611 phase c)	+0.0343 pu
Highest feeder voltage	1.0767 pu (675 phase b)	1.0781 pu (675 phase b)	+0.0014 pu
Phase voltages above 1.05 pu	17 of 50	17 of 50	+0
Phase voltages below 0.95 pu	0 of 50	0 of 50	+0
R1 (RG60 -> 632), largest phase current	587.7 A	284.9 A	-302.8 A
R2 (632 -> 671 at 671), largest phase current	470.0 A, forward	252.2 A, reverse	-217.8 A
Branch elements with reversed active power	0	7 (671-692 and the six segments of 632-671)	

1. Direction of the power flow

Base case (DG disconnected): active power and largest phase current per branch



DG connected: active power and largest phase current per branch

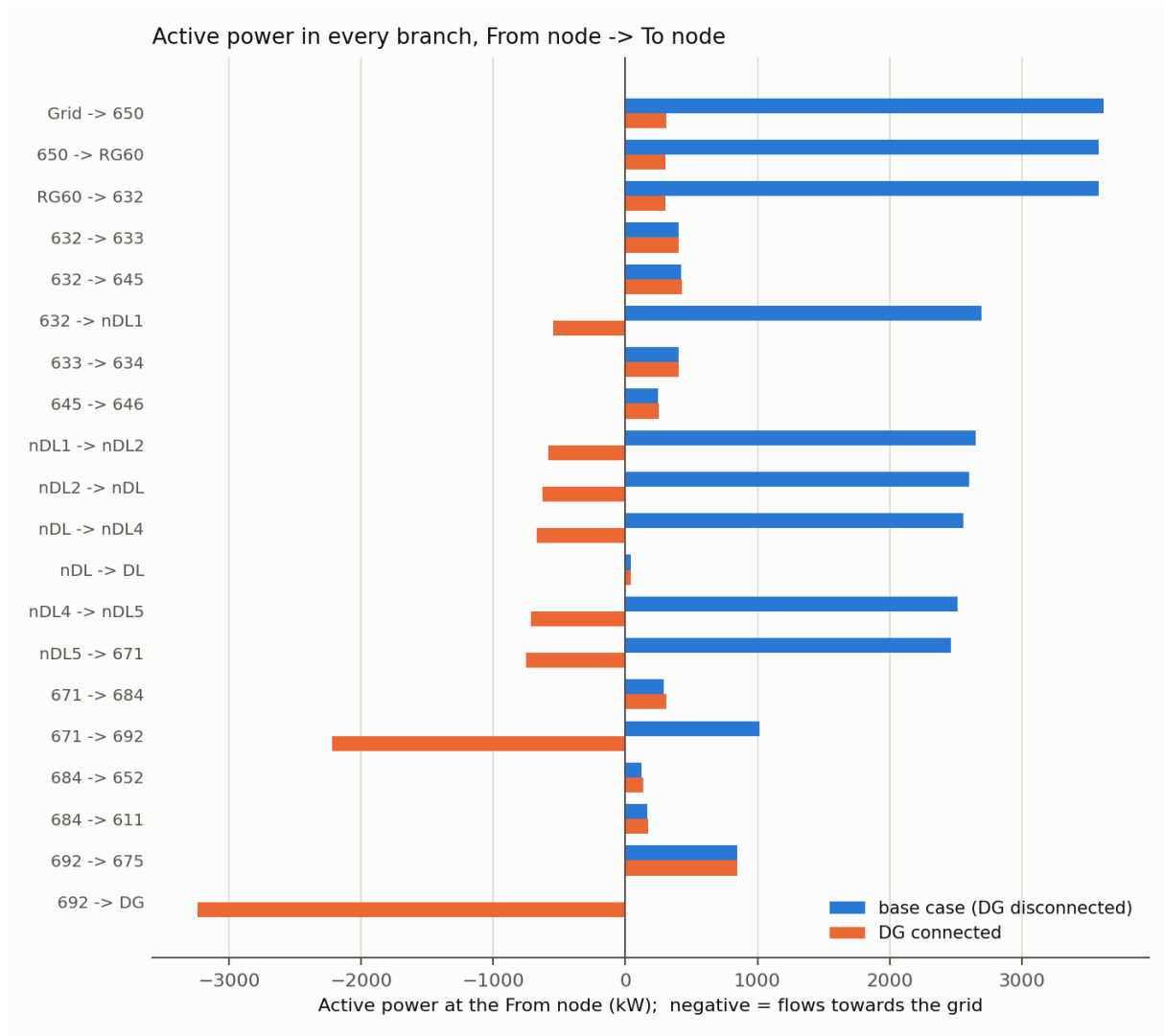


"mid" is the middle of line 632-671, where the distributed-load lateral DL is tapped; the line is modelled in six segments (label above mid: at 632; label below mid: at 671).

2. Branch flows, From node -> To node

From node	To node	P base (kW)	P with DG (kW)	Change (kW)	Q base (kvar)	Q with DG (kvar)	Flow, base	Flow, with DG
Grid	650	+3618.2	+310.3	-3307.9	+2003.3	+1900.6	Grid -> 650	Grid -> 650
650	RG60	+3583.0	+302.3	-3280.7	+1722.0	+1837.3	650 -> RG60	650 -> RG60
RG60	632	+3583.0	+302.3	-3280.7	+1722.0	+1837.3	RG60 -> 632	RG60 -> 632
632	633	+406.3	+406.1	-0.2	+301.1	+300.6	632 -> 633	632 -> 633
632	645	+420.9	+425.6	+4.7	+269.8	+272.5	632 -> 645	632 -> 645
632	nDL1	+2696.3	-543.3	-3239.6	+958.1	+1220.6	632 -> nDL1	nDL1 -> 632
633	634	+405.4	+405.2	-0.2	+299.9	+299.5	633 -> 634	633 -> 634
645	646	+247.6	+252.2	+4.6	+142.2	+144.9	645 -> 646	645 -> 646
nDL1	nDL2	+2649.8	-585.3	-3235.1	+914.5	+1191.8	nDL1 -> nDL2	nDL2 -> nDL1
nDL2	nDL	+2603.5	-627.3	-3230.8	+871.4	+1163.0	nDL2 -> nDL	nDL -> nDL2
nDL	nDL4	+2557.4	-669.3	-3226.7	+829.0	+1134.2	nDL -> nDL4	nDL4 -> nDL
nDL	DL	+40.0	+40.0	+0.0	+23.2	+23.2	nDL -> DL	nDL -> DL
nDL4	nDL5	+2511.5	-711.3	-3222.8	+787.0	+1105.4	nDL4 -> nDL5	nDL5 -> nDL4
nDL5	671	+2465.7	-753.3	-3219.0	+745.6	+1076.7	nDL5 -> 671	671 -> nDL5
671	684	+290.0	+307.6	+17.6	+67.1	+71.0	671 -> 684	671 -> 684
671	680	+0.0	+0.0	+0.0	-0.0	-0.0	no flow	no flow
671	692	+1015.1	-2217.8	-3232.9	+0.7	+340.1	671 -> 692	692 -> 671
684	652	+123.9	+135.7	+11.8	+82.9	+90.8	684 -> 652	684 -> 652
684	611	+165.5	+171.3	+5.8	-16.3	-20.3	684 -> 611	684 -> 611
692	675	+847.4	+847.0	-0.4	-148.3	-181.4	692 -> 675	692 -> 675
692	DG	+0.0	-3240.0	-3240.0	+0.0	+365.8	no flow	DG -> 692

Shaded: active power reversed by the DG. Values at the From node.



The DG's 3240 kW enter at 692, cover the loads at 692, 675, 671, 684, 611 and 652, and the rest (753 kW) flows back through 671 and the whole line 671 -> 632, where it supplies 633/634, 645/646 and the distributed load. The grid then only supplies 310 kW, but still 1901 kvar, because the DG delivers no reactive power in this operating point.

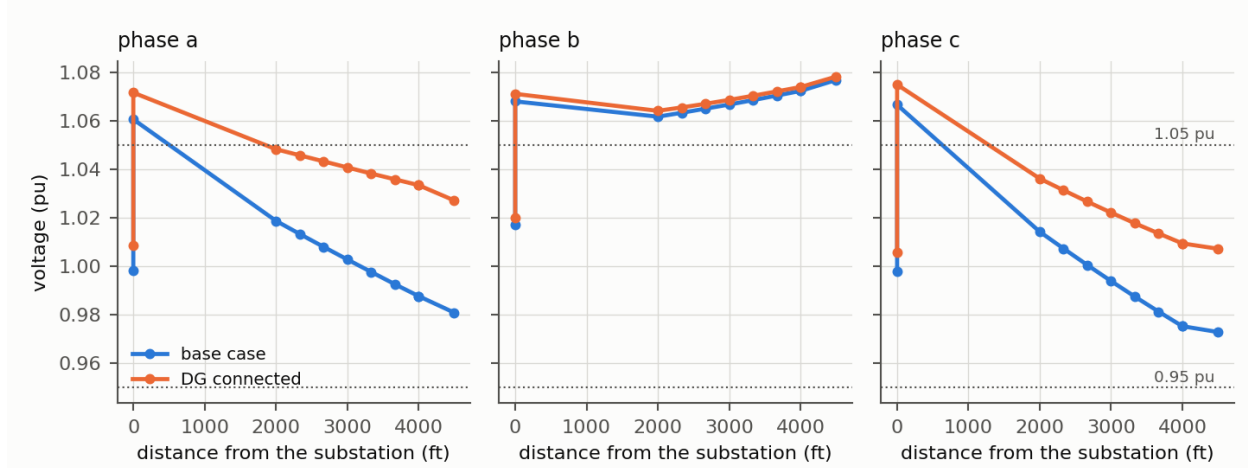
3. Branch currents per phase (A)

From node	To node	Ia base	Ib base	Ic base	Ia with DG	Ib with DG	Ic with DG	Largest phase: base -> DG (A)	Change
Grid	650	+18.6	+20.7	+23.3	-9.9	+8.2	+11.2	23.3 -> 11.2	-52 %
650	RG60	+592.7	+431.9	+628.1	+281.2	-195.1	+304.5	628.1 -> 304.5	-52 %
RG60	632	+557.9	+411.3	+587.7	+264.6	-185.8	+284.9	587.7 -> 284.9	-52 %
632	633	+81.0	+60.6	+62.8	+78.8	+60.3	+61.3	81.0 -> 78.8	-3 %
632	645	-	+143.1	+66.1	-	+143.2	+66.7	143.1 -> 143.2	+0 %
632	nDL1	+478.1	+212.2	+475.7	+210.3	-233.3	-179.3	478.1 -> 233.3	-51 %
633	634	+81.0	+60.6	+62.8	+78.8	+60.3	+61.3	81.0 -> 78.8	-3 %
645	646	-	+66.1	+66.1	-	+66.7	+66.7	66.1 -> 66.7	+1 %
nDL1	nDL2	+476.5	+206.5	+464.8	+209.5	-236.9	-174.2	476.5 -> 236.9	-50 %
nDL2	nDL	+474.9	+200.8	+453.8	+208.7	-240.6	-169.7	474.9 -> 240.6	-49 %
nDL	nDL4	+473.3	+195.2	+442.9	+207.9	-244.4	-165.9	473.3 -> 244.4	-48 %
nDL	DL	+1.6	+6.0	+11.3	+1.6	+5.9	+11.0	11.3 -> 11.0	-3 %
nDL4	nDL5	+471.6	+189.6	+431.8	-207.1	-248.2	-162.7	471.6 -> 248.2	-47 %
nDL5	671	+470.0	+184.0	+420.8	-206.3	-252.2	-160.2	470.0 -> 252.2	-46 %
671	684	+63.0	-	+71.1	+65.9	-	+71.3	71.1 -> 71.3	+0 %
671	680	0.0	0.0	0.0	0.0	0.0	0.0	0.0 -> 0.0	-
671	692	+229.7	+71.5	+178.8	-230.5	-392.8	-287.5	229.7 -> 392.8	+71 %
684	652	+63.0	-	-	+65.9	-	-	63.0 -> 65.9	+5 %
684	611	-	-	+71.1	-	-	+71.3	71.1 -> 71.3	+0 %
692	675	+205.9	+71.5	+124.5	+196.8	+71.6	+119.9	205.9 -> 196.8	-4 %
692	DG	0.0	0.0	0.0	-430.8	-421.7	-455.3	0.0 -> 455.3	-

The sign of a phase current is the direction of that phase's active power (- = towards the grid, shaded). "-" = the branch does not have that phase. With the DG the feeder is strongly unbalanced in direction: at R1 (RG60 -> 632) phase b carries power towards the grid while phases a and c still import.

4. Bus voltages (pu, phase-to-neutral)

Voltage along the main feeder 650 - RG60 - 632 - 671 - 692 - 675

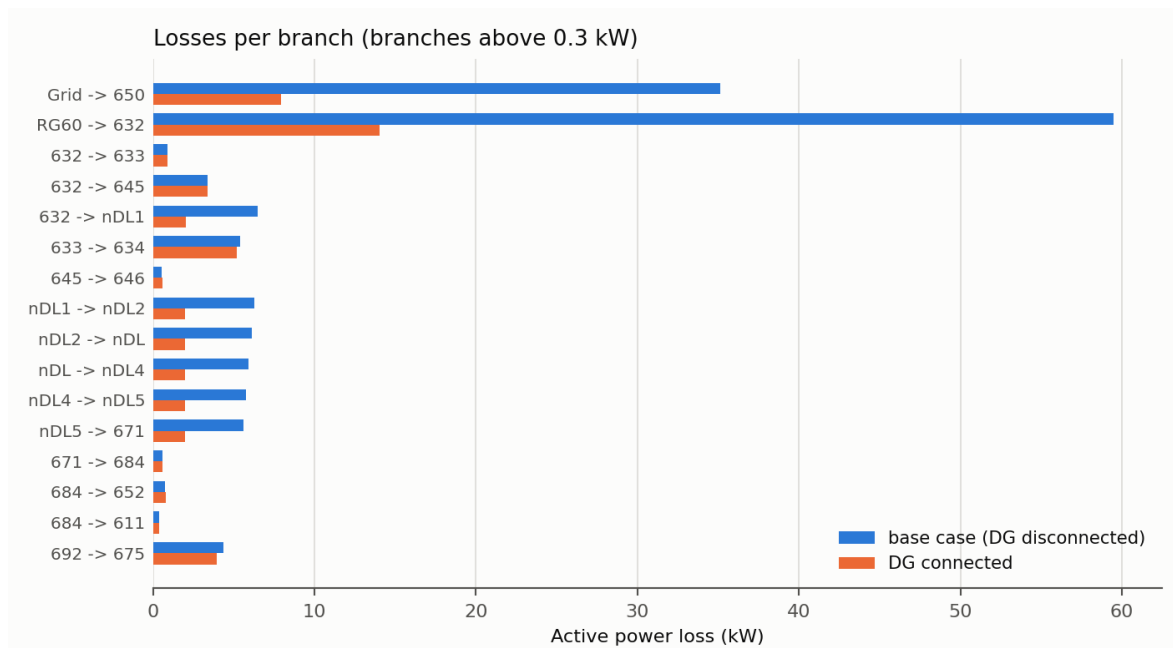


Bus	Va base	Va DG	dVa	Vb base	Vb DG	dVb	Vc base	Vc DG	dVc
650	0.9981	1.0084	+0.0103	1.0170	1.0200	+0.0030	0.9979	1.0057	+0.0078
RG60	1.0605	1.0715	+0.0110	1.0679	1.0710	+0.0031	1.0665	1.0748	+0.0083
632	1.0187	1.0482	+0.0295	1.0616	1.0640	+0.0024	1.0142	1.0361	+0.0219
633	1.0158	1.0454	+0.0296	1.0588	1.0611	+0.0023	1.0124	1.0344	+0.0220
645	-	-	-	1.0540	1.0563	+0.0023	1.0106	1.0325	+0.0219
nDL1	1.0133	1.0457	+0.0324	1.0632	1.0654	+0.0022	1.0073	1.0313	+0.0240
634	0.9918	1.0221	+0.0303	1.0406	1.0430	+0.0024	0.9936	1.0161	+0.0225
646	-	-	-	1.0522	1.0546	+0.0024	1.0085	1.0304	+0.0219
nDL2	1.0080	1.0432	+0.0352	1.0649	1.0669	+0.0020	1.0005	1.0266	+0.0261
nDL	1.0028	1.0407	+0.0379	1.0666	1.0685	+0.0019	0.9939	1.0221	+0.0282
nDL4	0.9977	1.0382	+0.0405	1.0684	1.0702	+0.0018	0.9875	1.0177	+0.0302
DL	1.0028	1.0407	+0.0379	1.0666	1.0685	+0.0019	0.9939	1.0221	+0.0282
nDL5	0.9926	1.0358	+0.0432	1.0703	1.0720	+0.0017	0.9813	1.0135	+0.0322
671	0.9877	1.0334	+0.0457	1.0722	1.0738	+0.0016	0.9753	1.0094	+0.0341
684	0.9859	1.0315	+0.0456	-	-	-	0.9732	1.0074	+0.0342
680	0.9877	1.0334	+0.0457	1.0722	1.0738	+0.0016	0.9753	1.0094	+0.0341
692	0.9877	1.0334	+0.0457	1.0722	1.0738	+0.0016	0.9753	1.0094	+0.0341
652	0.9811	1.0265	+0.0454	-	-	-	-	-	-
611	-	-	-	-	-	-	0.9712	1.0055	+0.0343
675	0.9809	1.0271	+0.0462	1.0767	1.0781	+0.0014	0.9729	1.0072	+0.0343
DG	1.0103	1.0302	+0.0199	1.0376	1.0390	+0.0014	0.9868	1.0269	+0.0401

Shaded yellow: above 1.05 pu. "-" = the bus does not have that phase. 634 is the 0.48 kV bus, DG the 0.69 kV generator bus.

The DG raises phases a and c along the whole feeder (up to +0.046 pu at the far end) and removes the voltage drop towards 671. Phase b is already above 1.05 pu in the base case (it is lightly loaded and the regulator taps are fixed), and the DG raises it slightly further.

5. Losses



From node	To node	Loss base (kW)	Loss with DG (kW)	Change (kW)
Grid	650	35.15	7.91	-27.24
RG60	632	59.50	14.02	-45.48
632	633	0.91	0.88	-0.03
632	645	3.38	3.39	+0.01
632	nDL1	6.47	2.02	-4.45
633	634	5.40	5.18	-0.22
645	646	0.56	0.57	+0.01
nDL1	nDL2	6.28	2.00	-4.28
nDL2	nDL	6.10	1.98	-4.12
nDL	nDL4	5.93	1.97	-3.96
nDL4	nDL5	5.76	1.98	-3.78
nDL5	671	5.61	1.98	-3.63
671	684	0.58	0.60	+0.02
684	652	0.72	0.79	+0.07
684	611	0.39	0.39	+0.00
692	675	4.38	3.96	-0.42
Total		147.1	49.6	-97.5

6. Loads and capacitors at the buses

Bus	Load base (kW / kvar)	Load with DG (kW / kvar)	Change (kW)	Capacitor base (kvar)	Capacitor with DG (kvar)
645	170.0 / 125.0	170.0 / 125.0	+0.0	-	-
nDL1	40.0 / 23.2	40.0 / 23.2	+0.0	-	-
634	400.0 / 290.0	400.0 / 290.0	+0.0	-	-
646	247.0 / 141.8	251.6 / 144.4	+4.6	-	-
nDL2	40.0 / 23.2	40.0 / 23.2	+0.0	-	-
nDL4	40.0 / 23.2	40.0 / 23.2	+0.0	-	-
DL	40.0 / 23.2	40.0 / 23.2	+0.0	-	-
nDL5	40.0 / 23.2	40.0 / 23.2	+0.0	-	-
671	1155.0 / 660.0	1155.0 / 660.0	+0.0	-	-
692	167.8 / 149.0	175.2 / 155.6	+7.4	-	-
652	123.2 / 82.8	134.9 / 90.6	+11.7	-	-
611	165.1 / 77.7	170.9 / 80.4	+5.8	94.3	101.1
675	843.0 / 462.0	843.0 / 462.0	+0.0	613.6	646.3

Constant-impedance and constant-current loads (646, 652, 692, 611) and the capacitors draw or supply more at the higher voltage with the DG, which is why the total load rises by 29.5 kW.

Findings

- **Power flow reverses on the main feeder.** With the DG, active power flows 692 -> 671 -> 632: through the 671-692 switch and all six segments of line 632-671 (7 branch elements). R2 (at the 671 end of line 632-671) carries 252.2 A in reverse instead of 470.0 A forward.
- **The grid import falls from 3618 kW to 310 kW**, and R1's largest phase current from 587.7 A to 284.9 A. The current at R1 does not fall in proportion, because the grid still supplies 1901 kvar.
- **Losses fall from 147.1 kW to 49.6 kW** (-66 %), mainly on RG60 -> 632, the substation transformer and line 632 -> 671.
- **Voltages rise.** The lowest feeder voltage goes from 0.9712 pu to 1.0055 pu. 17 of 50 phase voltages are above 1.05 pu in the base case and 17 with the DG; none is below 0.95 pu in either case.
- **Laterals are unaffected.** 632 -> 633 -> 634, 632 -> 645 -> 646, 671 -> 684 -> 611 / 652 and 692 -> 675 keep their direction and change only through the voltage dependence of their loads.